How to read our results

Tumor-aware MRI super-resolution: what the numbers mean, and how they can mislead

Companion to proposal.pdf · ACVSS Hackathon

The result, in one page

Sharpening a cheap blurry brain scan with AI can quietly delete a small tumor. It happens because the sharpening model is graded on average pixel error, and a small tumor is a tiny fraction of the pixels, so removing it barely costs the model anything on its own scorecard. We changed what the model is punished for, making errors inside the tumor cost far more, and then measured how much tumor survives.

It works. On 70 real BraTS patients the tumor-aware model loses far less tumor than the standard one, and it does so at the *same* segmentation quality, so it is not simply producing sharper pictures.

What the tumor detector was given	Tumor it missed	Caused by sharpening
The original scan, untouched	45.5%	—
Standard sharpening (baseline)	54.6%	+9.1 points
Tumor-aware sharpening (ours)	**48.1%**	+2.6 **points**

Read the right-hand column, not the middle one. The tumor detector we measure with already misses 45.5% of lesions on a perfect, untouched scan, so most of the middle column is not caused by sharpening at all. Sharpening adds 9.1 points of missed tumor on top of that floor, and **our objective removes 71% of that damage** (9.1 down to 2.6). Nearly every patient benefits: 66 of 70 improve, 1 is worse, and the effect is far outside chance ($p < 0.0001$).

Three things that make the finding credible, and one that limits it.

- It appears where the theory says it must and nowhere else. The benefit is there for the small enhancing tumor and *vanishes* for the whole tumor region, which is nine times larger. That is the prediction: the objective is only misaligned when the structure is too small to matter to a pixel-error score.
- Large tumors are essentially never lost by either model, so a raw figure near 50% should not be read as “half the tumors disappear”. The number is dominated by tiny fragments, which is what the term *lesion* below is about.
- Image quality is genuinely matched, not approximately: Dice 0.660 against 0.675 and brain-masked PSNR within 0.23 dB.
- **The limit:** it is not free. Protecting tumors makes the model readier to report one that is not there, and false alarms rise from 0.266 to 0.387. A second loss term recovers most of that cost, at the price of about half the benefit.

Terms

Three of these are used more narrowly here than in ordinary usage, and every confusing number in this document traces back to one of them.

Lesion	Clinically, an area of abnormal tissue: for us a tumor, one three-dimensional mass a radiologist would describe in a report. **In our metrics it means something much narrower**: one connected blob of the tumor mask within a *single 2D slice*. A scan is a stack of roughly 155 cross-sections, so one tumor crossing twenty of them counts as twenty lesions, a thin or broken enhancing rim fragments into several more within each slice, and a five-pixel speck counts the same as a whole tumor. This is why counts are large, why rates look alarming, and why lesions cannot be treated as independent samples.
Erasure	The failure we set out to fix. The reconstruction smooths a real tumor away, so the scan looks normal when it is not. Measured as the fraction of true lesions the downstream segmenter can no longer find. Also called pseudo-normalization. Lower is safer.
Hallucination	The opposite failure. The reconstruction invents tumor where there is none. Measured as the fraction of predicted lesions that overlap no true lesion. Lower is safer, and it tends to move in the opposite direction to erasure.
Floor	What the downstream segmenter misses when handed the *untouched original scan*, with no degradation and no reconstruction involved. It is large, so it is the denominator every erasure rate has to be read against.
Distortion-optimal	The standard approach, and our baseline: trained only to minimise average pixel error (PSNR).
Tumor-aware	Ours: the same network and data, trained with errors inside the tumor weighted far more heavily.
Dice	How well a predicted tumor outline overlaps the true one, from 0 to 1. Reported to show the two models are compared at equal segmentation quality.

Provenance

Figures come from the enhancing-tumor checkpoint trained on 17,233 axial slices from 468 BraTS patients (Medical Segmentation Decathlon brain task), split by patient, measured on the 70-patient validation split. Raw outputs are in results/ and the checkpoints are on Hugging Face at douyeszn/tumor-aware-sr. The proposal's results table comes from an earlier run on a smaller version of the dataset, so its digits differ.

Results

1. Why is the erasure rate around 50%? Is the model that bad?

No. Roughly nine tenths of that number is not caused by super-resolution at all.

The safety metric asks a downstream tumor segmenter to find the lesion, and then counts what it misses. So the number is a property of the whole pipeline, and the segmenter is part of the pipeline. If we hand that segmenter the *original, untouched scan*, with no degradation and no reconstruction anywhere in sight, it still misses **45.5%** of enhancing lesion components. That is its own blind spot, and it is the floor beneath every other number we report.

Image given to the segmenter	Lesions missed	Caused by our pipeline
Untouched original scan (the control)	45.5%	— (this is the floor)
Distortion-optimal reconstruction	54.6%	$+9.1$ points
Tumor-aware reconstruction (ours)	48.1%	$+2.6$ points

So the defensible claim is not "48.1% against 54.6%". It is that super-resolution introduces 9.1 points of erasure on top of the floor, and the tumor-aware objective removes 71% of that (9.1 down to 2.6). Quoting the absolute rates instead makes our own method look far worse than it is, and simultaneously hides the fact that the biggest weakness in the pipeline is the segmenter rather than the loss function.

2. What counts as a lesion? Are we losing whole tumors?

Almost never. Recall that a lesion here is one blob in one slice, so the count is dominated by very small ones. Split by area:

Lesion size	How many	Floor	Baseline adds	Ours adds
Small, < 50 px	6,762 (71%)	65.1%	$+13.5$	$+4.7$
Medium, 50–200 px	1,005	10.0%	$+6.4$	$+3.5$
Large, > 200 px	1,723	0.5%	$+0.7$	$+0.4$

Read the bottom row first. A lesion above 200 px is missed 0.5% of the time on a perfect scan and 0.9% after our reconstruction. Substantial tumors are not vanishing. The overall rate is high because 71% of all components are under 50 px, and two thirds of *those* are missed even on the original image.

Two honest caveats. Small does not mean unimportant, since a 30 px enhancing focus can be an early recurrence, and this is where the remaining work is. And the count is partly an artefact: a thin enhancing rim breaks into many separately counted pieces, and the metric weights a five-pixel speck the same as a whole tumor.

3. Why do we report per-patient numbers instead of per-lesion?

Because per-lesion numbers pretend to more precision than they have. One tumor spans many axial slices, and it appears as a separate "lesion" in each. Those outcomes are not independent: if a reconstruction loses that tumor, it loses it on every slice at once. Treating thousands of correlated records as thousands of independent samples makes the confidence interval look about 1.5 to 1.7 times tighter than it is.

So we compute one erasure rate per patient, pair the two objectives within each patient, and bootstrap by resampling *patients*. Reported that way the effect holds and the statement stays honest: on the larger dataset the mean paired difference is 6.5 points, 95% CI $[5.2, 8.2]$, with 66 of 70 patients improved and 1 worse.

4. What is the "hallucination penalty"?

The two safety failures pull in opposite directions. *Erasure* is missing a real tumor, and it is what we set out to reduce. *Hallucination* is the opposite mistake, reporting tumor where there is none, measured as the fraction of predicted lesion components that overlap no true lesion. The penalty is how much worse that gets when we switch from the standard objective to ours.

The mechanism is worth understanding, because it is not incidental. A lesion-weighted loss only asks that the tumor region be reconstructed accurately, so the cheapest way to satisfy it is to render anything tumor-like brighter and more solid, and that overshoot is exactly hallucination. Adding a term that requires the frozen segmenter's output to match the true mask punishes the mistake in both directions, and it takes the penalty from $+0.121$ down to $+0.017$ at the cost of roughly half the erasure gain. Which operating point is right is a clinical judgement, not ours: a screening service that cannot absorb false alarms should choose differently from one triaging known cases.